

One Machine, Many Molds

Running a HYDRA order across all mold slots on a single machine

Pattern: Meta-resource + Secondary resources + Cavity (NEST)

Modules: WRM · HLS · BDE · AIP

Source: MPDV HYDRA CUT-HDB data model (2021)

The problem

- One machine. It has **multiple slots**.
- Each slot holds **one mold**. Up to ~80 molds total.
- One production order should drive **all mounted molds at once**.

✗ Wrong: model each mold as a separate machine.

✓ Right: model the machine as a **meta-resource** that holds a list of mounted tools (molds).

Two shapes — don't confuse them

Shape	HYDRA functions	Your case?
N machines in parallel — one qty spread across separate machines	HLS-MFB, HLS-AGS, BDE-APF/SSG	X
1 machine, N mold slots — molds mounted in slots on one machine	HLS-BSR + WRM-NST	

This deck = the second shape.

Reality: 3 nested tiers (PU machine)

```
MACHINE (1, fixed physical slots)
├─ Order 1 = Product X → molds {...}
├─ Order 2 = Product Y → molds {...}
└─ Order 3 = Product Z → molds {...} ] concurrent (1..N)
```

- One machine runs **several orders at once** — each order = a different product.
- Each product mounts a **variable** number of molds.
- The machine has a **fixed** number of physical slots.

Two independent capacity caps

Cap	Field	Governs	Set to
Parallel orders	Available capacity [per mill]	OPs at once	N_orders × 1000
Parallel molds/slots	Anonymous resource Quantity	molds mounted at once	fixed slot count
Ceiling	Logon of several OPs = Y	MDE terminal max	≤ 20

Orders and slots are **separate levers** — that's the key insight.

Fixed slots = anonymous slot-pool

HLS capacity-checks the **tools/resources** on an operation, not just the machine.

- Model the slots as **one anonymous resource**, **Quantity = N** (fixed).
- Every operation lists it, **anzahl** = molds that product mounts.
- HLS enforces **Σ molds across all orders $\leq N$ slots**.

→ Works no matter how many orders or how many molds per product.
(needs secondary-resource capacity-check license)

Manual & Auto planning — both fit

- **Manual** drag: warns on double assignment, but allows it. Molds shown one under another.
- **Auto** (`graptsbap`): respects raised capacity + slot-pool; won't overload.

The "auto can't multi-assign" limit only applies at the **default 1000** capacity. Raise it → auto packs orders correctly.

Molds vary per product → **Log on with OP = Explicit** (operator scans mounted molds).

The mental model

```
ORDER (1 operation)
└─ PRIMARY: MACHINE      meta_res = 'J'    (type MNR)
    └─ slot / NEST 1 → mold A      \
    └─ slot / NEST 2 → mold B      | type WZ
    └─ ...                  | res_familie = MOLDPPOOL
    └─ slot / NEST 80 → mold ...  /
```

- Machine = **primary** resource (the meta-resource).
- Molds = **subordinate / secondary** resources hanging off it.
- Slot = **cavity (NEST)**.

Core idea in one line

The machine declares "I hold a list of tools."

The order asks for "N tools from the mold family" — not one fixed mold.

HYDRA binds the available molds to the order and books each slot.

Driven entirely by **resource master data** + **WRM-NST cavity config** — no custom code.

Implementation — Step 1 & 2

1. Machine = meta-resource (res_bestand)

Field (PDM ID)	Value
meta_res (RES.OPT:METARES)	J — "has resource list"
res type	MNR (machine)

2. Each mold = tool resource in one pool

Field (PDM ID)	Value
res type (res_ress_typen)	WZ / WNR (Werkzeug)
res_familie (RES.RESFAMID)	one family id for all ~80 molds

Implementation — Step 3 & 4

3. Define the slots — cavity management

Field	Meaning
<code>res_bestand.param_str_02</code> → <code>RES.TLGNEST</code> (WRM-NEST)	"Partitioning due to cavities" = how many slots active

4. Order needs molds from the pool — `res_bedarfszuord`

Field (PDM ID)	Role
<code>res_nr_m</code> (<code>RESBEDRES.RES:M</code>)	superordinate / required resource = the machine
<code>res_nr_t</code> (<code>RESBEDRES.RES:T</code>)	subordinate resource = the mold family

→ Order demands N tools from the family, not one fixed mold.

Implementation — Step 5 & 6

5. Auto-attach molds when the order logs on

res_bestand.mit_anmelden RES.OPT:AUTOANMELD	Behaviour
J	mold logs on with order (A_AN) / off (A_AB) automatically
E	explicit — operator scans the slot mounted (best for 80-mold pool)
N	never (DNC resources)

6. Allow simultaneous bookings

res_bestand.mehrfach RES.OPT:MULTIMNR	"can be logged on several times / simultan"
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Runtime — who occupies which slot

Table **res_ress_belegung** — *Zuordnung Auftrag zu Werkzeug*
Read by the HLS scheduling board.

Column	Use
belegungsart	A = order · S = lock · W = maintenance
ressource ress_typ	mold ID / WZ
anzahl	number of resources needed
bel_von_dat bel_bis_dat	maintenance / lock window per mold

Each of the 80 molds = its own occupancy row → board shows all slots filled.

Quantity & quality per slot

BDE-NBT — *Changed Partitioning Based on Cavities*

- Output qty + cycle time recompute when active cavity count changes.
- Pull one mold for repair → 79 active → targets adjust automatically.

AIP-NES — *Quality data relating to cavities*

- Captured per `nest_nr` (cavity number).
- Sample size = per-characteristic × number_of_cavities.
- Every mold's parts traceable to its slot.

Verify it's working (SQL pointers)

- Which molds occupied by an order? → `res_ress_belegung WHERE belegungsart='A' AND key = :auftrag_nr`
- Which order on machine now? → `auftrag_status.prod_kenn='L'` + `auftrags_bestand.masch_nr`
- Mold maintenance/lock windows? → `res_ress_belegung WHERE belegungsart IN ('S','W')`

See [\[\[HYDRA Order-Machine Query Pattern\]\]](#) for the full query set.

Rollout checklist

1. ☐ Set machine `meta_res='J'` , type `MNR`
2. ☐ Create 80 mold resources, type `WZ` , one `res_familie`
3. ☐ Configure cavities/slots via WRM-NST (`param_str_02` / `RES.TLGNEST`)
4. ☐ Link order tool demand → mold family (`res_bedarfszuord`)
5. ☐ Set `mit_anmelden` (`J` or `E`) + `mehrfach`
6. ☐ Enable BDE-NBT cavity partitioning + AIP-NES per-nest QC
7. ☐ Validate on HLS board (`res_ress_belegung` occupancy)

Config click-path (MOC + WRM)

1. **MOC** → **resources**: machine `MNR` (meta-resource), molds `WNR`, one **Family**
2. **WRM** → **Required resources**: molds → one pool per product (HYDRA picks at logon)
3. **Anonymous slot-pool**: `Quantity = N` = fixed physical slots
4. **Tool detail**: Original partitioning = `# cavities` · enable "**Partitioning due to cavities**"
5. **Operation tool list**: add pool (`anzahl` = molds) + **slot-pool** · "**Log on with OP = Explicit**"
6. **Machine config**: "**Logon of several OPs = Y**" (≤ 20) · **Available capacity = $N_{orders} \times 1000$**
7. **Planning**: manual drag **or** auto (`graptsbap`) — both honour both caps
8. **Runtime**: `AIP_M_TLG_NEST` cavity open/close · occupancy in `res_ress_belegung`

Full reference: [HYDRA Multi-Tool Resource Configuration]

Thank you

Full write-up: <wiki/questions/hydra-multi-mold-machine.md>

Module ref: [[HYDRA WRM Module]] · [[HYDRA HLS Module]] · [[HYDRA BDE Module]]

Source: [[hydra-cutldb-data-model]]